

National Tsing Hua University
11320IEEM 513600
Deep Learning and Industrial Applications
Homework 4

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Due on 2025/05/01.

Note: DO NOT exceed 3 pages.

1. (15 points) Experiment with different window sizes and steps. Train the model using 3 different combinations of window size and step. Evaluate the Mean Squared Error (MSE) for each configuration. Report the MSEs using a table and analyze the results. (Approximately 100 words.)

No.	Window sizes	Steps	MSE
1	10	15	281.5
2	15	20	612.7
3	5	10	138.5

2. (Approximately 200 words.)

- (i) (15 points) Include 'Volume' as an additional input feature in your model. Discuss the impact of incorporating 'Volume' on the model's performance.

設定(Window sizes, Steps) = (10, 15) , 加入" Volume" 的特徵後 , MSE 從281.5變成1335.3 , 模型的表現變差很多 , 從df.describe的值可以看到 , Volume的平均值遠大於其它欄 , 讓模型過於關注Volume的欄位 , 使模型變得不穩定 , 應做標準化。

- (ii) (15 points) Explore and report on the best combination of input features that yields the best MSE. Briefly describe the reasons of your attempts and analyze the final, optimal input combination.

從以下表格可以看到 , 在所有特徵都被選取的情況下 , MSE才是最小的 , 而且只用單一特徵預測是MSE是最高的 , 代表特徵間有交互作用存在 , 在這個情況下 , 當除了Volume的特徵以外都被選取的話 , 模型才是最好的。

Open	High	Low	Close	MSE
V	V	V	V	281.5

V	V	V		490.9
V	V		V	497.3
V		V	V	499.7
	V	V	V	490.7
V	V			680.1
V		V		569.5
V			V	643.0
	V	V		707.9
	V		V	625.5
		V	V	659.5
V				777.3
	V			758.3
		V		797.5
			V	789.3

3. (15 points) Analyze the performance of the model with and without normalized inputs in Lab 4. You can use experimental results or external references (which must be cited) to support your conclusions on whether normalization improves the model's performance. (Approximately 100 words.)

在2(i)曾提到，因為 Volume 的數值與其他特徵相差太大，需要做標準化，於是我在接受所有 5 個特徵(Open, High, Low, Close, Volume)當做 Input 並做標準化後，MSE 從原本的 1335.3 顯著下降成了 296.4；但是在原本使用的前 4 個特徵，做了標準化後，MSE 從 281.5 變成了 407.1，可見標準化並不保證提升模型表現。

4. (10 points) Why should the window size be less than the step size in Lab 4? Do you think this is correct? If you use external sources, please include references to support your response. (Approximately 50 words.)

Window size大於step會導致重疊的情況發生，在數據少時可以增加數據，但是在數據大時，太多重疊可能會讓數據變得很大，也可能會有一些冗餘的情況發生。

Reference: [machine learning - Any issue with "overlapping" sliding windows in time-series](#)

5. (15 points) Describe one method for data augmentation specifically applicable to time-series data. Cite references to support your findings. (Approximately 100 words.)

Decomposition-based series augmentation 像是 STL 或 Robust STL，將序列拆為trend、seasonal和residual三部分，然後對residual進行bootstrapping或其他變換，再與trend、seasonal重構新序列。此方法保留宏觀角度(Trend、seasonal)，同時在Noise方面(Residual)增加多樣性，提升預測和異常檢測性能。

Reference: [\[2002.12478\] Time Series Data Augmentation for Deep Learning: A Survey](#)

6. Discuss how to handle window size during inference in different model architectures (approximately 150 words):

通常是根據資料情況設定window size大小，一次需要看多少才會判斷的比較好，再來是硬體的限制，如果沒辦法一次輸入太多，就只能將window size輸入的小一點，那若是輸入長度變長時，以下不同模型會用一些方法做切割：

- (i) (5 points) Convolution-based models

可以用sliding-window的方式將序列較長的輸入切割。

- (ii) (5 points) Recurrent-based models

原本就能接受長序列，所以可以直接一次跑完整序列，利用內部hidden state傳遞。

- (iii) (5 points) Transformer-based model

長度會受訓練時設定，太長的序列需要chunk/sliding-window attention，並保留重疊區域讓資訊連續。